# DESCRIPTION OF COURSEWORK

| | |
|---|---|
| Course Code | EEE314 |
| Course Name | Digital System Design |
| Lecturer | Ir. Dr. Chua Ming Yam |
| Academic Session | 2023/09 |
| Assessment Title | Open-ended Final Project |

## A. Introduction/ Situation/ Background Information

FPGA stands for field-programmable gate array. It is an integrated circuit that implements code in hardware to execute a thousand times faster than in a processor. These circuits, or arrays, consist of configurable logic blocks (CLBs), memory, or other elements. FPGAs boast rapid growth over the past decade due to the wide range of applications they can handle. For example, it has become a major subject of research and utilization in many engineering fields like hardware acceleration, wireless body sensor networks, diagnostic imaging, electromedical segments, cardiac management, genomics/life science, computer architectures, and more. This course demonstrates how digital design and rapid prototyping could be facilitated by FPGAs and hardware description languages (Verilog).

## B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO2 Design solutions for complex engineering problems in digital design and FPGA

CLO3 Build digital systems using a modern EDA tool

## C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating

published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).

ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.

iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

**2.** All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

## D. Instruction to Students

1. This is a group final project with 2 – 3 members per group.
2. You are required to use Quartus Prime and DE10-Standard development boards to complete this final project.
3. Use the cover page provided for your report.
4. Submit the softcopy of your report and Quartus archived project file to Ir. Dr. Chua Ming Yam (mingyam.chua@xmu.edu.my)
5. Submission deadline: 29 Dec 2023, 11:59:59 pm.

## E. Evaluation Breakdown

| No. | Component Title | Percentage (%) |
|---|---|---|
| 1. | Quartus project (group) | 20 |
| 2. | Presentation (individual) and demonstration (group) | 20 |
| 3. | Project report (group) | 60 |
| | **TOTAL** | **100** |

**F. Task(s)**

You are required to design a communication controller using Pulse Width Modulation (PWM) to send and receive ASCII characters at the data rate of at least 9600 baud per second. You must implement the communication controller on the DE10-Standard development board and be flexible to use any onboard hardware. You are expected to validate your design using testbench and measurement equipment.

*Warning: This final project is an open-ended design problem with many solutions. A similar design/code/script will be considered plagiarism. Plagiarism of any kind will not be tolerated.*

Report guidelines:

- Font type: Times New Roman, Font size: 12pt.
- Line Spacing: 1.5
- Layout: Single-column page layout.
- Suggested report contents:
  - Introduction
  - Project background
  - Objectives
  - Methodology
  - Results and discussion
  - Learning outcome achievement
  - Conclusions and recommendations

**APPENDIX 1**

## MARKING RUBRICS

| **Component Title** | Quartus Project (Group) | | | | **Percentage (%)** | 20 | |
|---|---|---|---|---|---|---|---|
| **Criteria** | **Score and Descriptors** | | | | | **Weight (%)** | **Marks** |
| | **Excellent (5)** | **Good (4)** | **Satisfactory (3)** | **Inadequate (2)** | **Poor (1)** | | |
| Sub-modules (CLO3-PLO5-P4) | Well-working sub-modules with no issues. | Complete sub-modules but with 1 or 2 issue(s). | Complete sub-modules with > 2 issues. | Some sub-modules were missing. | Blank project / almost all sub-modules were missing. | 10 | |
| Entire projects (CLO3-PLO5-P4) | Well working complete project with no issues. | Complete project but with 1 or 2 issue(s). | Complete project but with > 2 issues. | Incomplete project. Major parts were missing. | Blank project / almost all parts were missing. | 10 | |
| | | | | | **TOTAL** | **20** | |

| **Component Title** | Presentation (Individual) and Demonstration (Group) | | | | **Percentage (%)** | 20 | |
|---|---|---|---|---|---|---|---|
| **Criteria** | **Score and Descriptors** | | | | | **Weight (%)** | **Marks** |
| | **Excellent (5)** | **Good (4)** | **Satisfactory (3)** | **Inadequate (2)** | **Poor (1)** | | |
| Technical content (Group) (CLO2-PLO3-C4) | Information on the slides is clear with pertinent facts and/or statistics that support the conclusions/ideas with evidence.<br><br>Closure summarizes important points leaving no doubt about the intended purpose. | Information on the slides is somewhat clear, with some examples, facts and/or statistics that support the topics.<br><br>Include some data or evidence that supports conclusions.<br><br>Makes the points intended and then summarizes the information to end the presentation. | The information on the slides is somewhat clear, but with weak facts and/or statistics that do not adequately support the topics.<br><br>Includes very thin data or evidence.<br><br>Makes the points and has a conclusion to end the presentation. | The information on the slides is somewhat lacking, with no support for topics, and presents insufficient support for ideas or conclusions.<br><br>The presentation is a series of random facts with no logic to the sequence of the presentation. | The information on the slides is very much lacking, with no support for topics, and presents insufficient support for ideas or conclusions. | 5 | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Presentation skills (Individual) (CLO2-PLO3-C4) | Clear voice and correct, precise pronunciation of terms.<br><br>Good eye contact and body language, steady rate, enthusiasm, and confidence. | The voice is clear and at a proper level. Most words are pronounced correctly.<br><br>Some eye contact and body language, steady rate, excessive rehearsal. | Occasional mispronunciation of terms.<br><br>Little eye contact and body language, uneven rate, only little expression. | Significant mumbling and incorrect pronunciation of terms<br><br>Monotonous, with no eye contact and body language, the rate of speech is too fast or too slow. | Not able to present the thought | 5 | |
| Q&A (Individual) (CLO2-PLO3-C4) | Thorough, well-elaborated, and persuasive answers were provided.<br><br>Impressive composure as a professional speaker when taking and answering questions. | Clear and concise answers were provided.<br><br>Calm, clear, and confident when taking and answering questions. | Partly appropriate answers to questions.<br><br>Attentive when taking and answering questions. | Out-of-topic or evasive answers to questions.<br><br>Visibly downcast or distracted during the Q&A session. | Not able to answer the question | 5 | |
| Project demonstration (Group) (CLO3-PLO5-P4) | Successfully show all output results and elaborate the project outcome excellently | Show almost all the output results and elaborate on the project outcome sufficiently | Show almost all the output results but fairly elaborate on the project outcome | Show the output results partially and fairly elaborate on the project outcome | Fail to show the output results and unable to elaborate on the project outcome | 5 | |
| | | | | | **TOTAL** | **20** | |

| **Component Title** | Project Report (Group) | | | | **Percentage (%)** | 60 | |
|---|---|---|---|---|---|---|---|
| **Criteria** | **Score and Descriptors** | | | | | **Weight (%)** | **Marks** |
| | **Excellent (5)** | **Good (4)** | **Satisfactory (3)** | **Inadequate (2)** | **Poor (1)** | | |
| Project background & objectives (CLO2-PLO3-C4) | Well-written and concise background of the study with a clear definition of | A good background that includes some definitions of key terms and engineering concepts. | The background study is relevant but does not define key terms and engineering concepts. | The background study is partially relevant.<br><br>Poorly defined objectives. | The background study is irrelevant.<br><br>The objectives are not relevant. | 5 | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | key terms and engineering concepts.<br><br>The objectives are clear and well-defined. | The objectives are clear. | Somewhat well-defined objectives. | | | | |
| Description of the proposed system (CLO2-PLO3-C4) | A good description of the proposed system. | A somewhat good description of the proposed system. | A satisfactory description of the proposed system | A fair description of the proposed system. | A poor description / no description of the proposed system. | 10 | |
| System complexity & modularization (CLO2-PLO3-C4) | Involve complex tasks with complex system integration. | Involves complex tasks with a significant amount of system integration | Involves complex tasks with some system integration. | Involves simple tasks with little system integration. | Involves very simple tasks with no system integration (very direct). | 10 | |
| Presentation of findings/results (Simulation & measurement) (CLO3-PLO5-P4) | Clear and comprehensive presentation of results/findings. | Good presentation of results/findings. | Results/findings were presented but incomprehensive. | Results/findings were presented but inadequate. | No presentation of findings/results. | 10 | |
| Discussion (CLO2-PLO3-C4) | Details and outputs are justified and labeled clearly. The findings are discussed excellently concerning theory with sufficient evidence. | Outputs are explained in detail and labeled clearly. The findings are discussed concerning theory and demonstrate appropriate evidence. | Most outputs are explained and labeled sufficiently. The findings are discussed concerning theory and demonstrate appropriate evidence. | Details and outputs are fairly explained and labeled. The findings are fairly discussed concerning theory and show partially appropriate evidence. | Details and outputs are not explained and labeled clearly. The findings are not discussed concerning theory and inappropriate evidence. | 10 | |
| Learning outcome achievement (CLO2-PLO3-C4) | A well-written LO achievement highlighting important points. | A good write-up of LO achievement but requires minor improvement | A satisfactory write-up for LO achievement. | A write-up of LO achievement that lacks relevant points. | A poor write-up of LO achievement. | 5 | |
| Conclusion (CLO2-PLO3-C4) | Conclusion covers all the important points and directly relates the findings to the objectives. | Conclusion covers most of the important points and adequate relation of the findings to the objectives. | Conclusion on major points are drawn, but less value of the relation of the findings to the objectives. | Inappropriate conclusion with little or no relation of the finding to the objectives. | Failed to write the conclusion. | 5 | |
| Organization, language, and formatting (CLO2-PLO3-C4) | Very well-organized report.<br><br>Excellent written | Well-organized report.<br><br>Well-written | The report is organized.<br><br>Appropriate | Inconsistent format such as font type, size, and spacing. | Unorganized report.<br><br>Poor use of grammar and language. | 5 | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | grammar without errors.<br><br>The format is consistent, figures and tables are neatly done. | grammar.<br><br>The format is generally consistent, figures and tables are neatly done. | written grammar.<br><br>Mostly consistent format of figures and tables with some spelling mistakes. | Figures and tables are disorganized and several grammar mistakes. | | | |
| | | | | | **TOTAL** | **60** | |